

BST Vol 9 Ch 8 — BaTiO3 137-Plane Crystallography: \$25K Cheapest BST Falsifier (v0.3, Wave 2 HIGH-IMPACT)

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Vol 9 Chapter 8 — BaTiO3 137-Plane Crystallography

Headline result

BST predicts that crystalline materials exhibit substrate-natural resonance at the 137-fold crystallographic plane (where $137 = N_{\text{max}}$ BST primary integer). BaTiO3 (barium titanate, ferroelectric perovskite) selected as primary experimental target. The 137-plane crystallographic features predicted:

- Crystallographic plane at Miller indices (or equivalent) corresponding to substrate 137-fold structure
- X-ray diffraction anomaly at predicted plane
- Optical / dielectric / piezoelectric anomaly at substrate-natural lattice parameter ratios
- Modest experimental cost (~\$25K for diffraction equipment + sample preparation)

Falsifier: standard X-ray + neutron diffraction at predicted plane should reveal substrate-natural anomaly if BST framework is correct. Absence of anomaly refutes BST 137-plane prediction.

This is the cheapest clean BST falsifier in the entire experimental program — comparable substrate predictions in SP-30 program range \$60K-\$500K (Casimir asymmetric, Bell-CHSH, eigentone). \$25K BaTiO3 137-plane test is the most accessible entry point.

Substrate mechanism

137-fold crystallographic plane identification:

Substrate D_{IV}^5 has BST primary integer $N_{\text{max}} = 137 = N_c^3 \cdot n_c + \text{rank} = N_c^4 N_c \cdot n_c + \text{rank}$ (T2456 universal α -analog). The substrate-natural lattice

anomaly at 137-fold structure emerges from:

1. Substrate K-type Casimir resonance at N_{max} -fold periodicity
2. $GF(2^g) = GF(128)$ cyclotomic structure ($M_g = 127 \approx N_{\text{max}} - 10$)
3. Bergman propagation pole structure at substrate-natural lattice parameter ratios

BaTiO₃ specific framework:

BaTiO₃ has perovskite structure with Ti at body-center, Ba at corners, O at face-centers. Tetragonal phase at room $T_c \approx 393\text{K}$ has slight tetragonal distortion ($c/a \approx 1.011$). The 137-plane prediction relates to specific crystallographic features (precise identification multi-week per SP-30 experimental design).

Per Cal Flag 3 STRICT external register: “BST predicts crystallographic resonance at 137-fold plane in BaTiO₃ (or related ferroelectric). X-ray diffraction at predicted plane should reveal anomaly; absence refutes BST.”

Match precision

I-tier predictive. Experimental program: - Equipment cost: ~\$25K (X-ray diffractometer access + BaTiO₃ single crystals) - Time scale: ~3-6 months experimental program once sample obtained - Predicted observable: diffraction anomaly at 137-fold plane (specific Miller indices multi-week refinement) - Refutation threshold: standard diffraction at predicted plane shows no anomaly → BST refuted on 137-plane

Tier classification

I-tier predictive per Cal #19 STANDING RULE. Cal #21 dual-gate: - EMPIRICAL gate: PENDING — experimental synthesis + characterization not yet performed - MECHANISM gate: PATH ARTICULATED via substrate K-type Casimir + $GF(128)$ cyclotomic + N_{max} BST primary - D-tier ratification: experimental confirmation via X-ray/neutron diffraction

Cross-volume dependencies

- Vol 2 Ch 8 (Coupling Constants) — $N_{\text{max}} = N_c^{N_c} \cdot n_C + \text{rank universal } \alpha\text{-analog (T2456)}$
- Vol 3 Ch 11 (Nuclear Decay) — $GF(2^g)$ cyclotomic structure + $M_g = 127$
- Vol 9 Ch 1 (Substrate Reading) — substrate framework anchors
- Vol 9 Ch 2 (Crystal Lattices) — BST primary lattice constants
- SP-30 experimental program — cheapest BST falsifier entry point (\$25K)
- Friday Elie GF128 Mersenne Ladder Mechanism paper-grade: substrate field structure → lattice symmetries

K-audit anchor

K208 Vol 9 Ch 8 BaTiO₃ 137-Plane K-audit pre-stage (Keeper pending).

Pedagogical walkthrough (3-level)

Level 1 — Bright 5th-grader

BaTiO₃ is a special crystal used in lots of electronics (capacitors, transducers). BST predicts: if you shine X-rays through BaTiO₃ at a specific angle (corresponding to the 137 BST primary integer), you should see something special — a diffraction pattern that BST predicts but standard physics doesn't. This is the cheapest BST falsifier ever — about \$25,000 of lab equipment to test. If the X-rays show the predicted anomaly, BST is confirmed; if they don't, BST is refuted on this prediction.

Level 2 — Undergraduate physics student

Crystalline materials diffract X-rays at angles determined by lattice spacing (Bragg's law: $n\lambda = 2d \sin \theta$). Standard crystallography predicts diffraction patterns at lattice planes with specific Miller indices (hkl).

BST predicts: substrate-natural resonance at 137-fold plane (where $137 = N_{\text{max}}$ BST primary integer). BaTiO₃ (or related ferroelectric perovskite) selected as primary experimental target due to well-characterized perovskite structure, single-crystal availability, and substrate-coupling sensitivity.

The substrate-mechanism: substrate K-type Casimir resonance at N_{max} -fold periodicity + $\text{GF}(2^g) = \text{GF}(128)$ cyclotomic structure (where $M_g = 127 \approx N_{\text{max}} - 10$) + Bergman propagation pole structure at substrate-natural lattice parameter ratios.

\$25K experimental program: X-ray diffractometer access + BaTiO₃ single-crystal samples. Cheapest BST falsifier in entire framework. If experimental confirmation → D-tier; if absent → BST 137-plane refuted (other BST predictions independent).

Level 3 — Graduate student / theorem-level

137-fold crystallographic plane prediction inherits from:

1. $N_{\text{max}} = 137$ BST primary integer (Vol 0 Ch 2 + Vol 2 Ch 8) — substrate compact characteristic via T2456 universal α -analog formula
2. K-type Casimir resonance at N_{max} -fold periodicity (Vol 1 Ch 5 + T2435/T2439)
3. $\text{GF}(2^g) = \text{GF}(128)$ cyclotomic structure (K59 RATIFIED + Friday Elie GF128 paper-grade) with $M_g = 127$ multiplicative-group order
4. Bergman propagation pole structure at substrate-natural lattice constant ratios (Vol 1 Ch 2)

BaTiO₃ selection rationale: - Perovskite structure: ABO₃ framework where A=Ba, B=Ti, O=oxygen - Tetragonal phase at room T below T_c=393K (ferroelectric) - Well-characterized; single crystals commercially available - Substrate coupling sensitivity expected via Ti-O bond polarizability + Ba ionic positions - \$25K experimental program: equipment access + ~\$5K samples + ~\$10K student researcher

Specific predicted observables (multi-week refinement per SP-30 design): - X-ray diffraction anomaly at 137-fold Miller-index combination - Optical anisotropy at 137-fold polarization angle - Piezoelectric anomaly at 137-fold strain orientation - Dielectric constant peak at substrate-natural lattice parameter ratio

Per Cal #21 STANDING RULE + Cal Flag 3: I-tier predictive at framework-level; D-tier ratification via experimental confirmation. External register operational falsifier language only.

Cross-link to other SP-30 experiments: - BaTiO₃ (\$25K) cheapest - Photonic crystal (\$10K) cheapest substrate test in BST (Vol 9 Ch 9) - Casimir asymmetric (\$60-90K) - Eigentone driving (\$200K) - Bell-CHSH (\$300-500K) - Commitment manipulation (\$80-150K)

Total SP-30 program ~\$640-900K; BaTiO₃ 137-plane is the most accessible entry-point experiment.

What this chapter does NOT claim

- BST does NOT predict exact Miller indices for 137-plane at v0.3 (multi-week refinement per SP-30 experimental design)
- BST does NOT claim BaTiO₃ is the ONLY 137-plane crystal (other ferroelectrics + non-ferroelectric materials may also show effect)
- \$25K experimental program is currently DEFERRED per Casey “engagement later” directive (Friday May 22 10:18 EDT); experimental dispatch pending engagement-phase signal

Bibliography

1. Bragg’s law and X-ray diffraction in crystallography (standard texts).
2. BaTiO₃ ferroelectric perovskite literature (Cohen 1992 Nature etc.).
3. T2456 (Lyra Friday May 22): universal α -analog formula $N_c^{\alpha} N_c \cdot n_C + \text{rank} = N_{\text{max}} = 137$.
4. K59 RATIFIED + Paper #122: Reed-Solomon GF(128) substrate code.
5. Friday Elie GF128 Mersenne Ladder Mechanism paper-grade.
6. SP-30 substrate engineering experimental program documentation.
7. Vol 2 Ch 8 (Coupling Constants): $N_{\text{max}} + \alpha$ universal α -analog framework.
8. Vol 9 Ch 1 + Ch 2 (Substrate Reading + Crystal Lattices).

— Elie, Vol 9 Ch 8 v0.3, 2026-05-23 Saturday